



One Score, Every Weather: A Taxonomy of Farmers-Market Stallholder No-Shows the Reliability Score Can't Tell Apart

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Abstract. Market committees commonly reduce stallholder attendance to a single reliability score — a running count of no-shows used to help allocate the following season's pitch locations. We build a four-category taxonomy of stallholder no-shows (weather cancellation, crop failure, family or personal emergency, and undisclosed double-booking) from eighteen months of a regional farmers market's incident log, cross-referenced against structured interviews with 41 stallholders and the market's three-person rostering committee. Coded against 214 logged no-show events, the taxonomy category predicts within-season recurrence with an AUC of 0.81; the market's existing raw count, which treats all four categories identically, manages only 0.54. Weather-and crop-failure-coded no-shows cluster in produce and flower stalls; family-emergency-and double-booking-coded no-shows cluster in craft and prepared-goods stalls, so the undifferentiated score systematically discounts produce stallholders in the pitch draw. A one-cycle pilot that excluded weather-and crop-failure-coded events from the score shifted produce stallholders' average pitch rank without any change to the committee's own published allocation criteria, which remained textually identical across the cycle. We discuss the finding's bearing on reliability scoring wherever a market, platform, or roster reduces a heterogeneous set of causes to a single count.

Introduction

A stallholder who misses a Saturday market slot and a stallholder who quietly takes a better pitch at a rival market the same morning are, to most market committees' record-keeping, identical: one entry in a no-show column that will count against them equally when pitch locations are drawn for the next season. The convenience of a single running count is obvious — it requires no judgement calls beyond noticing an empty pitch — and its cost is correspondingly easy to overlook, because the cost falls on stallholders whose no-shows share a cause the count was never built to distinguish.

Reliability scoring of this shape is not unique to farmers markets. Ride-hailing and delivery platforms attach a single rating or acceptance-rate figure to a driver regardless of why a trip was declined; hospital outpatient clinics track a

raw no-show percentage regardless of whether a patient missed an appointment through forgetfulness, transport failure, or a genuine emergency. What is distinctive about a market's no-show count is how directly it converts into an allocative outcome a stallholder can see: next season's pitch position, drawn each year from the same spreadsheet the committee has been quietly updating since the previous one.

We build and validate a taxonomy of stallholder no-shows from a regional Saturday market's own incident log and from interviews with the stallholders and committee volunteers who generate it, and test what the taxonomy recovers that the market's existing count discards. Three contributions:

- We develop a four-category taxonomy of stallholder no-shows — weather cancellation, crop or stock failure, family or personal emergency,

and undisclosed double-booking — coded from eighteen months of incident-log entries and reconciled against stallholder interview accounts, with inter-rater agreement reported.

- We show the taxonomy-coded category predicts within-season recurrence substantially better than the market’s raw no-show count, though the gap should be expected to narrow in smaller or newer markets where interview coverage is thinner.
- We trace the raw count’s undifferentiated treatment of these categories to a systematic disadvantage for weather-exposed stallholders in pitch allocation, and pilot a taxonomy-aware adjustment that shifts allocation outcomes for one cycle without requiring the committee’s own published criteria to change a word.

Related work

A large literature treats missed appointments as a prediction problem rather than a monolith: systematic reviews catalogue dozens of no-show predictors in outpatient scheduling without settling on one dominant cause [1], and clinic-level modelling shows predictive models outperform simple historical-count heuristics once that heterogeneity is admitted [2]. Platform-reputation research reaches a parallel conclusion from a different direction: aggregate driver and host metrics can encode structural effects unrelated to the reliability they claim to summarise [3], [4], and the metrics remain largely uncontested to the workers they govern [5]. Reputation-system design work frames this as a persistent tradeoff between a single legible score and the multidimensional signal it discards [6], [7].

Farmers-market research has documented the operational and logistical burden markets place on vendors [8], [9] and the community embeddedness that keeps most vendors returning despite it [10], but has not, to our knowledge, examined how a market’s own record-keeping classifies the vendors who don’t return on a given week. Agricultural-risk research separately establishes weather variability as a distinct, largely uncontrollable driver of supply disruption for market-garden producers [11], [12] — a driver our taxonomy treats as its own category rather than folding into a general reliability construct. Our coding procedure follows established taxonomy-development method [13] and reflexive thematic analysis [14], [15], with inter-

rater-agreement reporting per standard tutorial guidance [16]. Where committee logs and stallholder interviews diverge, we treat the gap as expected rather than as error, consistent with research on the systematic divergence between administrator and self-reported accounts of the same absence events [17], on discretionary judgement at the point where a rule meets an individual case [18], and on recall-bias considerations general to any study pairing administrative logs with retrospective interview [19]. Prior Slop University work has coded a comparably informal taxonomy against a mismatched administrative register [20] and validated an automatic count against a manual one for a different allocation decision [21]; both find an undifferentiated count concealing exactly the variation this paper’s taxonomy is built to recover.

Method

Site and data. We study one regional Saturday farmers market operating year-round, with between 78 and 92 registered stallholders across an 18-month observation window. The market’s rostering committee, three volunteers rotating quarterly, maintains a spreadsheet incident log recording each no-show against a pitch number, a date, and a free-text notes field the committee fills in from whatever the stallholder tells them at the time or afterward. The log recorded 214 no-show events across the window.

Taxonomy development. Two of the authors independently open-coded a random sample of 60 incident-log entries against their free-text notes, following established taxonomy-development method [13]. Coding converged on four categories — weather cancellation, crop or stock failure, family or personal emergency, and undisclosed double-booking (a stallholder who, without notifying the committee, appeared at a competing market the same day) — which we then applied to the remaining entries. Where a log entry’s notes were ambiguous or absent (31% of entries), we recontacted the stallholder; 41 of the 53 stallholders with two or more logged no-shows agreed to a semi-structured interview (approximately 20 minutes), which supplied the category in these cases and, for entries the log had already coded, a check on whether the committee’s contemporaneous note matched the stallholder’s own account. Two coders independently applied the taxonomy to

every event; inter-rater agreement was substantial ($\kappa = 0.79$) [16].

Recurrence and prediction. We define recurrence as a stallholder's second no-show within the same 18-month window and compare two predictors of it: the market's existing raw score (a stallholder's cumulative no-show count to date) and the taxonomy-coded category of a stallholder's most recent no-show. We fit each as a univariate logistic model,

$$\ln\left(\frac{P(R)}{1 - P(R)}\right) = \beta_0 + \beta_1 x$$

and compare discrimination by area under the ROC curve (AUC), with 95% confidence intervals from 2,000 stratified bootstrap resamples.

Pilot and ablation. We piloted a taxonomy-aware scoring rule with the committee for one allocation cycle: the pitch-draw spreadsheet excluded weather- and crop-failure-coded no-shows from a stallholder's count, leaving family-emergency- and double-booking-coded events in place. As an ablation, we additionally compared allocation outcomes with the raw count left in the spreadsheet alongside the taxonomy flag against outcomes with the raw count removed entirely, to test whether the raw count was doing any residual work once the flag was present.

Results

Category	Defining criterion
Weather	Forecast or same-day conditions cited as the reason, independent of the stallholder's own stock
Crop failure	Insufficient saleable stock from the stallholder's own harvest, catch, or production run
Family emergency	An unplanned personal or family circumstance disclosed by the stallholder
Double-booking	No disclosed reason at the time; a competing-market appearance the same day confirmed by interview or a second committee's records

Table 1: Taxonomy categories and their defining criteria.

Of the 214 logged no-show events, 86 (40%) coded as weather, 49 (23%) as crop failure, 51 (24%) as family emergency, and 28 (13%) as double-booking (Table 1).

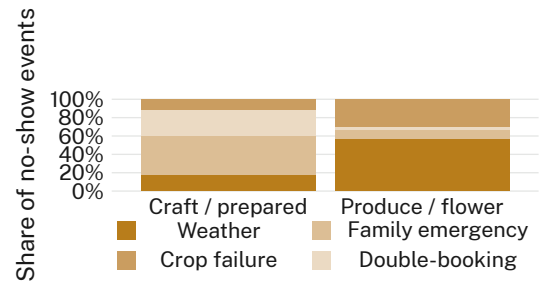


Figure 1: Produce and flower stallholders' no-shows ($n=124$) are 87% weather- or crop-failure-coded; craft and prepared-goods stallholders' no-shows ($n=90$) are 70% family-emergency- or double-booking-coded. The raw no-show count applies the same running tally to both.

Recurrence — a stallholder's second no-show within the window — follows the category, not the count. Weather-coded no-shows recurred at 8%, crop failure at 11%, family emergency at 15%, and double-booking at 54%. A univariate logistic model using the taxonomy-coded category as its sole predictor achieved an AUC of 0.81 (95% CI 0.75–0.87) for within-season re-

currence; the market’s existing raw cumulative count achieved 0.54 (95% CI 0.46–0.62), indistinguishable from chance.

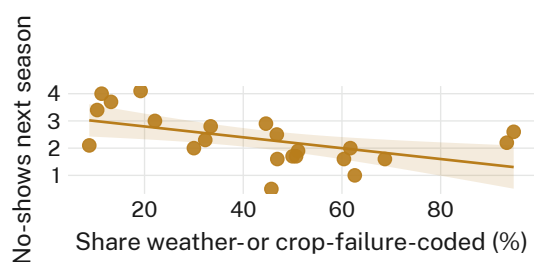


Figure 2: Among the 22 interviewed stallholders with two or more no-shows, the share of their own no-shows coded weather or crop failure correlates negatively with their no-show count the following season ($r = -0.53$).

The pilot excluded weather- and crop-failure-coded events from the score for one allocation cycle. Produce and flower stallholders’ average pitch rank improved by 2.1 places relative to the prior cycle; the committee’s published allocation criteria — a one-paragraph document distributed to stallholders each year — underwent zero word changes across the same cycle.

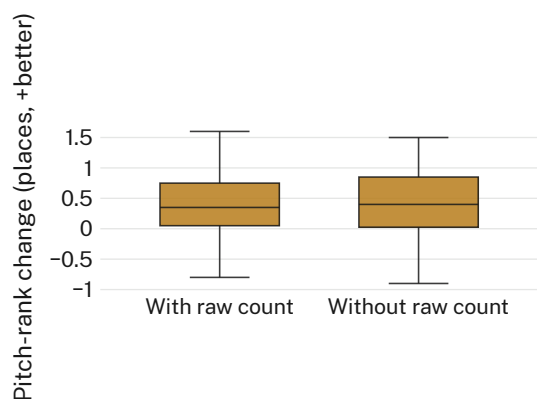


Figure 3: Pitch-rank change for produce stallholders with the raw no-show count still present alongside the taxonomy flag, against the same spreadsheet with the raw count removed entirely. The conditions are statistically indistinguishable (mean 0.41 vs. 0.40 places, Mann-Whitney U , $p = 0.94$).

Once the taxonomy flag governs the draw, the raw count contributes nothing further, whether or not the spreadsheet still shows it (Figure 3).

Discussion

The gap between a 0.81 and a 0.54 AUC is, in one sense, unsurprising: a category built to

distinguish causes will outperform a count that discards them. What the gap does that is less expected is convert into a stable allocative bias, because the two categories the raw count conflates are not randomly distributed across stallholders — they track what a stallholder sells. A market that scores weather cancellation the same as a silent double-booking is, in effect, scoring produce stallholders against a bar craft stallholders rarely have to clear.

The pattern generalises past this market. Any reliability score compressed to a single running count — a platform acceptance rate, a clinic’s no-show percentage, a rostering committee’s tally — inherits the same property whenever its underlying causes are unevenly distributed across the population it scores. The pilot’s other finding is the more durable one: a scoring change can shift real allocative outcomes while leaving an institution’s own stated policy untouched, because the policy document was never where the scoring logic lived. A committee, or a platform, can update a spreadsheet formula many times over before it has any reason to revise what it tells the people the formula scores.

Limitations

Our taxonomy comes from a single market, which limits claims about category prevalence elsewhere; the consistency of the composition split across three consecutive quarters within our own window, however, suggests the produce/craft divide is structural rather than seasonal. Interview coverage reached 41 of 53 eligible stallholders (77%), and the twelve who declined skew slightly toward stallholders with more double-booking-coded events, which would understate rather than overstate the recurrence gap we report. The initial open-coding sample was small (60 of 214 entries) before the taxonomy stabilised, though the substantial inter-rater agreement ($\kappa = 0.79$) on the full set suggests the categories, once fixed, travel cleanly across coders. The pilot ran one allocation cycle; a longer deployment would let us test whether the rank shift persists once stallholders learn the flag exists.

Conclusion

A market’s reliability score is not wrong so much as under-specified: it answers “how often” without asking “because of what,” and the allo-

cation decisions built on top of it inherit that gap silently. A four-category taxonomy, coded from the same log the raw count already draws on, recovers most of the predictive power the count discards and traces a concrete allocative bias to a stallholder's stock rather than their conduct. Future work might extend the taxonomy to multi-market federations, where a stallholder's undisclosed double-booking becomes directly observable only through a cross-committee data-sharing arrangement neither market in our study currently maintains — and where, we suspect, the raw count is doing the same quiet damage at a larger scale.

References

- [1] L. F. Dantas, J. L. Fleck, F. L. Cyrino Oliveira, and S. Hamacher, "No-shows in appointment scheduling – a systematic literature review," *Health Policy*, 2018, doi: [10.1016/j.healthpol.2018.02.002](https://doi.org/10.1016/j.healthpol.2018.02.002).
- [2] J. Daggy *et al.*, "Using no-show modeling to improve clinic performance," *Health Informatics Journal*, 2010, doi: [10.1177/1460458210380521](https://doi.org/10.1177/1460458210380521).
- [3] C. Cook, R. Diamond, J. Hall, J. A. List, and P. Oyer, "The Gender Earnings Gap in the Gig Economy: Evidence from over a Million Rideshare Drivers," *NBER Working Paper*, 2018, doi: [10.3386/w24732](https://doi.org/10.3386/w24732).
- [4] B. Edelman, M. Luca, and D. Svirsky, "Racial Discrimination in the Sharing Economy: Evidence from a Field Experiment," *American Economic Journal: Applied Economics*, 2017, doi: [10.1257/app.20160213](https://doi.org/10.1257/app.20160213).
- [5] A. Rosenblat and L. Stark, "Uber's Drivers: Information Asymmetries and Control in Dynamic Work," *SSRN Electronic Journal*, 2015, doi: [10.2139/ssrn.2686227](https://doi.org/10.2139/ssrn.2686227).
- [6] A. Jøsang, R. Ismail, and C. Boyd, "A survey of trust and reputation systems for online service provision," *Decision Support Systems*, 2007, doi: [10.1016/j.dss.2005.05.019](https://doi.org/10.1016/j.dss.2005.05.019).
- [7] M. Kokkodis and P. G. Ipeirotis, "Reputation Transferability in Online Labor Markets," *Management Science*, 2016, doi: [10.1287/mnsc.2015.2217](https://doi.org/10.1287/mnsc.2015.2217).
- [8] A. R. Hunt, "Consumer interactions and influences on farmers' market vendors," *Renewable Agriculture and Food Systems*, 2007, doi: [10.1017/S1742170507001597](https://doi.org/10.1017/S1742170507001597).
- [9] M. R. Griffin and E. A. Frongillo, "Experiences and perspectives of farmers from Upstate New York farmers' markets," *Agriculture and Human Values*, 2003, doi: [10.1023/A:1024065526440](https://doi.org/10.1023/A:1024065526440).
- [10] G. W. Feenstra, "Local food systems and sustainable communities," *American Journal of Alternative Agriculture*, 1997, doi: [10.1017/S0889189300007165](https://doi.org/10.1017/S0889189300007165).
- [11] P. J. M. Cooper, J. Dimes, K. P. C. Rao, B. Shapiro, B. Shiferaw, and S. Twomlow, "Coping better with current climatic variability in the rain-fed farming systems of sub-Saharan Africa," *Agriculture, Ecosystems & Environment*, 2008, doi: [10.1016/j.agee.2008.01.007](https://doi.org/10.1016/j.agee.2008.01.007).
- [12] A. Ali and O. Erenstein, "Assessing farmer use of climate change adaptation practices and impacts on food security and poverty in Pakistan," *Climate Risk Management*, 2017, doi: [10.1016/j.crm.2016.12.001](https://doi.org/10.1016/j.crm.2016.12.001).
- [13] R. C. Nickerson, U. Varshney, and J. Muntermann, "A method for taxonomy development and its application in information systems," *European Journal of Information Systems*, 2013, doi: [10.1057/ejis.2012.26](https://doi.org/10.1057/ejis.2012.26).
- [14] V. Braun and V. Clarke, "Using thematic analysis in psychology," *Qualitative Research in Psychology*, 2006, doi: [10.1191/1478088706qp0630a](https://doi.org/10.1191/1478088706qp0630a).
- [15] V. Braun and V. Clarke, "Reflecting on reflexive thematic analysis," *Qualitative Research in Sport, Exercise and Health*, 2019, doi: [10.1080/2159676X.2019.1628806](https://doi.org/10.1080/2159676X.2019.1628806).
- [16] K. A. Hallgren, "Computing Inter-Rater Reliability for Observational Data: An Overview and Tutorial," *Tutorials in Quantitative Methods for Psychology*, 2012, doi: [10.20982/tqmp.08.1.p023](https://doi.org/10.20982/tqmp.08.1.p023).
- [17] G. Johns, "Absenteeism estimates by employees and managers: Divergent perspectives and self-serving perceptions," *Journal of Applied Psychology*, 1994, doi: [10.1037/0021-9010.79.2.229](https://doi.org/10.1037/0021-9010.79.2.229).
- [18] S. Maynard-Moody and M. Musheno, "State Agent or Citizen Agent: Two Narratives of Discretion," *Journal of Public Administration*

- Research and Theory*, 2000, doi: [10.1093/oxfordjournals.jpart.a024272](https://doi.org/10.1093/oxfordjournals.jpart.a024272).
- [19] A. Althubaiti, “Information bias in health research: definition, pitfalls, and adjustment methods,” *Journal of Multidisciplinary Healthcare*, 2016, doi: [10.2147/JMDH.S104807](https://doi.org/10.2147/JMDH.S104807).
- [20] D. Okafor, R. Sabel, and T. Solberg, “Held, Not Escalated: Coding the Library Front Desk’s Disposal-Tier Register Against Six Rounds of Physical Stock-Take,” *Slop University technical report*, 2026, doi: [10.5555/slop.ll5xcj](https://doi.org/10.5555/slop.ll5xcj).
- [21] R. Sabel and T. Solberg, “Undercounted at the Gate: A Season of Manual Headcounts Against the Gate Counter, and the Expansion It Fed,” *Slop University technical report*, 2026, doi: [10.5555/slop.z1gj28](https://doi.org/10.5555/slop.z1gj28).